Phantom study of a self-shielded X-ray bone age assessment instrument against scattered radiation in children

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Abstract

Hand-wrist radiography is the most common and accurate method for evaluating children's bone age. To reduce the scattered radiation of radiosensitive organs in bone age assessment, we designed a small X-ray instrument with radioprotection function by adding metal enclosure for X-ray shielding. We used a phantom operator to compare the scattered radiation doses received by sensitive organs under three different protection scenarios (proposed instrument, radiation personal protective equipment, no protection). The proposed instrument showed greater reduction in the mean dose of a single exposure compared with radiation personal protective equipment especially on the left side which was proximal to the X-ray machine (≥80.0% in eye and thyroid, ≥99.9% in breast and gonad). The proposed instrument provides a new pathway towards more convenient and efficient radioprotection.

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Keywords

- Bone age
- Protective device
- Radiation exposure
- Children
- Phantoms

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